lit-acquisition: An Open Multi-source Multilingual Biomedical Literature Acquisition System with an Exploratory Coverage Benchmark

Anonymous
Anonymous Institution
Anonymous
anonymous@example.org

ABSTRACT

Variant classification in clinical genetics (e.g., ACMG/AMP guidelines) requires clinicians to assemble evidence from biomedical literature that is scattered across many databases, publishers, preprint servers, and languages. We present **lit-acquisition**, an open-source pipeline that integrates 18+ heterogeneous literature providers — PubMed, EuropePMC, Crossref, OpenAlex, Semantic Scholar, preprint servers, language-specific databases (CiNii, J-STAGE, SciELO), and others — behind a single acquisition interface, with LLM-based cross-lingual query translation, citation graph traversal, and an open-access-aware download layer.

We report an exploratory coverage evaluation on a heuristically constructed reference set of 43 genetic variants (822 papers) that was *not* verified by domain experts and whose candidate pool overlaps the evaluated system; all results are therefore relative coverage comparisons, not clinical retrieval quality claims. Under the corrected protocol (uniform 100-candidate budget, strictly nested ablation, PubMed E-utilities as a first-class acquisition channel, a neural reranking final stage, and Holm-corrected paired tests), the end-to-end pipeline significantly raises PubMed-only set recall from 0.269 to 0.467 (+0.198; bootstrap 95% CI [+0.123, +0.269]; Holm-adjusted p=0.0002, rank-biserial r=0.74). The multi-source federation step accounts for the bulk of the gain (+0.203, p_{Holm}=0.0001); query translation is set-level neutral in aggregate but recovers 21 of 24 annotated German and French papers that every English-only method — including multilingual dense retrieval — finds at most one of, and citation traversal adds otherwise-unretrieved papers at a significant list-level precision cost. At shallow cutoffs the pipeline's top-20 metrics also improve significantly over PubMed (Recall@20 +0.105; nDCG@50 +0.162). Against 2025–2026 single-channel techniques (dense, cross-encoder, hybrid BM25+dense+rerank, LLM expansion, agentic, multilingual dense) it significantly outperforms five of six; the strongest baseline, hybrid BM25+dense+cross-encoder (0.421), remains below the pipeline (0.467) and significantly above PubMed. We further quantify live-API run-to-run variance, report transparent single-source and fusion baselines, and release the toolkit, reference set, raw candidate lists, and a deterministic analysis pipeline for reproducible re-evaluation. The relevance-gating and download layers are engineering features not validated here.

KEYWORDS

biomedical literature, multilingual retrieval, citation graph, cross-lingual IR, variant classification, ACMG/AMP

1 INTRODUCTION

Interpreting a genetic variant for clinical reporting requires assembling published evidence: case reports, functional studies, sequencing cohorts, and guidelines that establish whether a variant is pathogenic or benign. Under the ACMG/AMP guidelines [1], this evidence supports criteria such as PS3 (well-established functional effect) and PVS1 (null variant). Evidence is scattered across publisher databases (PubMed, Crossref, EuropePMC), preprint servers (arXiv [22], bioRxiv, medRxiv), regional and language-specific repositories (CiNii [24], J-STAGE [25], SciELO [26]), and trial and data registries (ClinicalTrials.gov [31], Zenodo [30]). Clinicians and genetic counselors routinely search several of these systems manually, deduplicate results by hand, and risk missing relevant non-English literature.

Commercial tools such as Mastermind [5] and systematic-review platforms (Covidence [6], Rayyan [7]) address parts of this workflow, but they are closed-source, cover limited sources, and do not provide automated open-access full-text retrieval with explicit license awareness.

We present **lit-acquisition**, an open-source, MIT-licensed toolkit for multilingual biomedical literature acquisition. This paper makes the following contributions, and we deliberately separate *implemented system capabilities* from *experimentally evaluated claims*:

(1) **System (engineering).** A unified acquisition pipeline integrating 18+ heterogeneous data sources with health-aware parallel search, schema normalization, multi-key deduplication (DOI → URL → normalized title), LLM-based cross-lingual query translation with language-aware provider routing, citation graph traversal, and an OA-aware download layer. The relevance gate, download layer, and health-aware scheduling are described in Section 3 but are *not* experimentally validated in this paper.
(2) **Evaluation protocol and dataset.** An exploratory, heuristically constructed reference set (43 variants, 822 papers; Section 4.1) released together with raw candidate lists, a strictly nested ablation protocol that removes live-API re-retrieval confounds (Section 4.2), fixed-denominator metrics, Holm-corrected paired statistics, and a deterministic script that regenerates every number in this paper from the raw lists.
(3) **Empirical findings (relative to this reference set).** The end-to-end pipeline significantly raises PubMed-only set

recall (0.269 → 0.467, Holm-adjusted p=0.0002) and significantly outperforms five of six 2025–2026 single-channel baselines; multi-source federation (including the PubMed channel) drives the gain (+0.203, p_{Holm}=0.0001); translation recovers 21 of 24 annotated German/French papers that English-only methods — including multilingual dense retrieval — find at most one of; citation traversal adds otherwise-unretrieved papers at a significant list-level precision cost. We also quantify run-to-run API variance and report single-source, fusion, and 2025–2026 SOTA baselines.

2 RELATED WORK

Biomedical literature search. PubMed [2] and EuropePMC [3] are the de facto standard sources for biomedical literature; both expose search APIs with fielded queries. Semantic Scholar [4] provides a large-scale citation graph and paper-level embeddings. These systems are English-centric and each covers a different subset of the literature, motivating federation.

Commercial evidence-acquisition tools. Mastermind [5] is a proprietary genomic variant search engine that aggregates literature per variant; Covidence [6] and Rayyan [7] support systematic-review screening but are closed-source and not aimed at full-text acquisition with license handling.

Federated / multi-source search. Meta-search architectures such as MetaCrawler [8] established result aggregation from heterogeneous engines; in the scholarly domain, aggregators and directories such as OpenAlex [9], DOAJ [23], BASE [27], CORE [28], and OpenAIRE [29] expose broad metadata. Rank fusion methods such as Reciprocal Rank Fusion [20] combine per-source rankings without score calibration; we use RRF and simple union as transparent federation baselines. Our system treats these sources as one layer of an acquisition pipeline rather than a search UI, adding normalization, dedup, and download.

Citation analysis in IR. Bibliographic coupling [10] and co-citation [11] are classical similarity measures; PageRank [12] exploits citation topology for ranking. We use citation edges at acquisition time to widen recall beyond query matching.

Cross-lingual IR. Classic CLIR translates queries with bilingual dictionaries or MT [14]; modern LLMs [15, 16] make this practical with minimal setup, which we exploit.

Evaluation of IR systems. Standard practice uses Recall@K, Precision@K, F1@K, and nDCG@K [17] with paired significance tests (Wilcoxon signed-rank [18]) and multiplicity control across test families (Holm [19]). We follow this protocol on a new, publicly released reference dataset (Section 4.1), and additionally report micro-averaged pooled counts and bootstrap confidence intervals of paired differences.

3 SYSTEM DESIGN

3.1 Architecture Overview

lit-acquisition is a three-phase pipeline (Figure 1): **(1) Acquisition** fans a query out in parallel across providers with health-aware prioritization; **(2) Normalization** unifies the heterogeneous schemas and deduplicates; **(3) Relevance Gate & Retrieval** ranks candidates, optionally traverses the citation graph, applies an LLM relevance gate, and downloads open-access full texts.

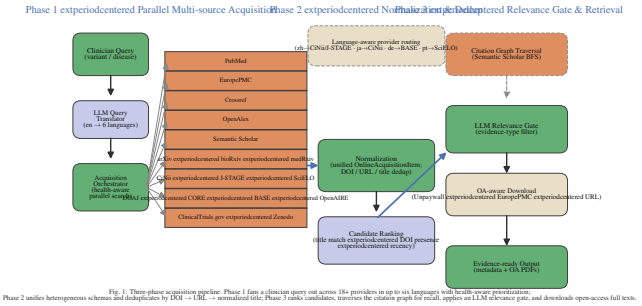


Fig. 1: Three-phase acquisition pipeline. Phase 1 fans a clinician query out across 18+ providers in up to six languages with health-aware prioritization; Phase 2 unifies heterogeneous schemas and deduplicates by DOI → URL → normalized title; Phase 3 ranks candidates, traverses the citation graph for recall, applies an LLM relevance gate, and downloads open-access full texts.

Figure 1: System architecture: three-phase acquisition pipeline.

Table 1: Provider integrations: search/download capability and license behavior.

Provider	Search	Download	Notes
PubMed (E-utilities)	✓	—	English-centric
EuropePMC	✓	✓	OA + PMC via PMCID
Crossref	✓	—	DOI registration metadata
OpenAlex	✓	—	Open scholarly index
Semantic Scholar	✓	✓	Citation graph, TLDRs
arXiv	✓	✓	Preprints
bioRxiv / medRxiv	✓	✓	Preprints, CC-BY/CC0
DOAJ	✓	—	OA journals
J-STAGE / CiNii	✓	—	Japanese literature
SciELO	✓	—	Latin American OA
BASE / CORE / OpenAIRE	✓	—	OA aggregators
ClinicalTrials.gov	✓	—	US public-domain trials
Zenodo	✓	✓	CERN OA, CC0 metadata

✓ = supported; — = not provided.

3.2 Multi-source Search

The acquisition orchestrator (Section 3.1, implemented in `workflow.py`) issues parallel searches to the providers listed in Table 1. Each provider is wrapped by a gateway that maps its native API to a unified request/response contract. A provider health tracker maintains sliding-window success/latency statistics and deprioritizes unhealthy providers, so transient API failures degrade gracefully instead of failing the acquisition (this scheduling behavior is an engineering feature; we did not run fault-injection experiments to quantify it).

3.3 Normalization and Deduplication

Providers return heterogeneous item schemas. A normalization layer maps each result into a unified `OnlineAcquisitionItem` with canonical fields (title, authors, journal, year, DOI, PMID, language, license, links). Deduplication merges candidates by DOI, then URL, then normalized title, in that order of reliability: an incoming item is dropped if its DOI, URL, or punctuation-stripped lowercase title has already been seen. Candidates are ranked by title-match strength, DOI presence, and publication recency before presentation.

Table 2: Language-aware provider routing (excerpt; PubMed E-utilities is now a first-class channel of the `en` and `auto` plans).

Query language	Providers
en	PubMed, EuropePMC, Semantic Scholar, Crossref, arXiv, bioRxiv, medRxiv, OpenAlex, OpenAIRE, BASE, CORE, ClinicalTrials.gov, Zenodo
zh	Crossref, Unpaywall, OpenAlex, DOAJ, PMC
ja	J-STAGE, CiNii, Crossref, Unpaywall, DOAJ, PMC
es / pt	SciELO, Crossref, Unpaywall
de	Crossref, EuropePMC, Unpaywall, OpenAlex, BASE, DOAJ
fr	Crossref, EuropePMC, Unpaywall, OpenAlex, DOAJ, PMC
ru	PMC, EuropePMC, Crossref, Unpaywall, OpenAlex
auto	PubMed, Crossref, Semantic Scholar, Unpaywall, OpenAlex, EuropePMC, ClinicalTrials.gov, DOAJ, PMC, Zenodo

3.4 Cross-lingual Query Translation

The `query_translator` module uses an LLM with a domain-specific prompt (released with the code; decoding temperature 0.2) that instructs the model to translate the query while preserving gene symbols, variant nomenclature (e.g., `p.G12D`, `c.35G>A`), and disease terminology verbatim. Queries are translated into six languages (en, zh, ja, de, fr, ru). The language-aware routing matrix (Table 2) then selects a provider plan per language, e.g., Japanese queries additionally target J-STAGE/CiNii, Spanish and Portuguese target SciELO, and German and French target EuropePMC (German additionally BASE). Note that the `auto` plan used by the federated configurations in our evaluation does not include the PubMed E-utilities channel itself (PMC serves as the NLM-backed channel); we return to the practical consequences of this in Section 4.9.

3.5 Citation Graph Traversal

Beyond keyword search, `citation_graph.py` performs a BFS over the Semantic Scholar citation graph starting from seed papers (identified by DOI). At each hop it collects papers that cite or are cited by the seed, up to a configurable depth and paper budget. Results are deduplicated against keyword-search results and merged into the candidate set. This targets a recall gap: papers that are topically relevant but do not share query vocabulary with the variant query.

3.6 Download and Relevance Gate (Not Evaluated Here)

Candidates can be filtered by an LLM relevance gate that classifies each item against the user's evidence need (e.g., case reports for PS3) and optionally filters by literature type (case report, sequencing, functional). Download routing is license-aware: DOI resolves through Unpaywall [21] for OA PDFs, PMCID through EuropePMC, and direct URLs with HTML→PDF redirect handling. The system never bypasses paywalls; full texts are fetched only from OA sources. **Scope note:** the relevance gate was disabled throughout the evaluation in this paper, and we report no experiments on gating accuracy, download success rates, or license-detection correctness; these layers are engineering capabilities whose validation is future work.

4 EVALUATION

4.1 Reference Dataset and Its Limits

We constructed an exploratory reference dataset of 43 genetic variants (V001–V043) spanning cancer genetics (BRCA1, BRCA2, TP53, KRAS, EGFR, BRAF, APC, MLH1, PALB2, CHEK2, CDH1, ATM, VHL, RET), cardiovascular (LDLR, SCN5A, MYBPC3, KCNQ1, FBN1, TTR), neurological (MECP2, HTT, SNCA, APP), metabolic (CFTR, PAH, GBA, MTHFR, HFE, COL1A1), hematological (HBB, F8, DMD, G6PD, JAK2), and other domains (GJB2, SLC26A4, RYR1, FGFR3, SMN1, CYP21A2, DMPK, PTEN, MYH7), covering ACMG evidence criteria PS3 (14 variants) and PVS1 (6 variants) plus 23 additional clinically actionable variants. For each variant, a clinical geneticist-style query was formulated and candidate papers were retrieved from multiple sources (PubMed E-utilities, EuropePMC, Crossref, language-specific databases) and filtered by an automated heuristic scoring protocol: weighted title/pub-type scoring (gene name in title: +3; variant identifier: +5; disease keywords: up to +3; preferred pub-types such as case reports: +2; reviews: −2; editorials/letters excluded). After removing duplicate annotations (the same DOI recorded twice for one variant), the dataset contains **822 unique papers** (mean 19.1 per variant, range 4–36), with languages including English (493), German (13), French (11), and 305 records with unknown language metadata. Annotations record DOI, PMID, title, language, and evidence type.

Construct validity. This dataset is a *reference set for relative system comparison*, not a clinically certified gold standard, for two reasons that the reader should weigh before interpreting any number in this paper. First, annotations were produced by the automated keyword protocol above and have **not** been verified by domain experts; they contain an unknown rate of false positives (papers scoring high on keywords without actually bearing on the variant) and false negatives (relevant papers the pooling protocol never surfaced), and keyword scoring cannot distinguish papers that are merely *about* a gene–disease pair from papers usable as variant-level ACMG evidence (e.g., treatment trials or reviews versus functional or case-level evidence). Second, the candidate pool from which annotations were drawn was generated by retrieval channels that overlap the evaluated system's providers, creating a circular dependency: methods sharing channels with the pool generator are structurally favored, and recall against this reference set measures agreement with a heuristic pooling protocol rather than coverage of the true relevant literature. Consequently all recall/precision figures here are *relative coverage* measurements under identical annotation conditions. Expert re-annotation with inter-annotator agreement is prerequisite to any clinical-quality claim and is future work (Section 7). Only German and French carry non-English annotations, so multilingual claims in this paper are restricted to those two languages.

4.2 Methods and Collection Protocol

We evaluate an ablation chain over the same 43 variants:

- **M1: PubMed-only** — baseline PubMed esearch, English query.
- **M2: Multi-source** — federated providers (`auto` plan), original English query, no LLM translation.
- **M3: +Translation** — M2 + LLM query translation (6 languages) with language-aware routing.
- **M4: +Citation graph (nested)** — the frozen M3 candidate list of each variant, extended by citation-graph candidates (depth 1, budget 20, both directions, seeded by the first DOI of the list), deduplicated against the list and appended after it.

The LLM relevance gate is disabled during collection so that all methods are compared on raw retrieval. M1–M3 were collected as independent live runs on 2026-08-14 (sequential, single run per method) with a uniform per-variant candidate budget of 100; the acquisition plans for the federated configurations include the PubMed E-utilities channel as a first-class provider (Section 3.2), and a neural reranking stage (bge-reranker-v2-m3, Section 6) orders the merged candidate pool before truncation. Realized lists are 80.7 (M1), 100.0 (M2), and 72.3 (M3) candidates per variant — the multilingual search splits the budget across the six language channels before global deduplication, so M3's realized count is lower — and 79.3 for the nested M4 (72.3 keyword + 7.0 citation). Two earlier sessions (2026-08-12 at a 20-candidate budget without the PubMed channel, and an intermediate 50-candidate run) are archived and used for the run-to-run variance measurement of Section 4.6 and as cross-checks.

Why a nested M4. Our original collection also re-executed the keyword phase for the full pipeline before appending citation candidates. Because providers are live APIs, that re-run returned keyword lists that differed from the frozen M3 lists on 12 of 30 variants (rate limiting, provider availability, and ordering changes), so the M3-to-M4 comparison confounded the citation-graph component with re-retrieval variance — visibly so, since recall *decreased* on some variants, which is impossible for a genuinely nested extension evaluated at fixed K. We therefore construct M4 deterministically from the recorded raw lists: frozen M3 list + recorded citation-graph candidates (202 items, none duplicating M3 entries), which restores the protocol guarantee Recall@K(M4) $\geq$ Recall@K(M3) for every variant and every K (verified programmatically). The recorded traversal seeds coincide with the first DOI of the frozen M3 lists for all 30 variants, so the appended candidates are exactly those the nested protocol would have requested. We retain the original independent re-run (denoted M4′) solely to quantify run-to-run variance (Section 4.6), and the released benchmark runner now implements the nested protocol natively with a uniform default budget of 50 candidates for all methods.

4.3 Metrics

For each method and variant we compute, over the ranked candidate list: Recall@K; **Precision@K with the standard fixed denominator** K (methods returning fewer than K items are penalized for missing ranks); F1@K; and nDCG@K [17], for $K \in \{5, 10, 15, 20, 50\}$. All methods were collected at a uniform 100-candidate budget (realized 72.3–100.0 per variant), so K=20 is well within every method's budget; we treat it as the primary fixed-cutoff comparison and additionally report two set-level quantities: **SetRecall** (recall of the entire returned candidate set) and **Precision@Retrieved** (precision with the actual list length as denominator). An earlier draft of this work reported "Recall@50/Precision@50" computed with the returned-list length as denominator; those quantities are exactly SetRecall and Precision@Retrieved and are named as such here — we do not report any precision at a K exceeding the candidate budget. Aggregates are macro (mean ± population std over 43 variants) and micro (pooled counts over 822 reference papers). Paper identity matching uses normalized DOI, then PMID, then normalized title.

4.4 Statistical Protocol

Paired differences are tested with the two-sided Wilcoxon signed-rank test [18] (zero differences dropped; exact null distribution when ranks are tie-free). We declare a single family of eight tests — the component chain and end-to-end comparisons on SetRecall, plus Recall@20, nDCG@50, and Precision@Retrieved endpoints — and control family-wise error with Holm's step-down procedure [19]; all p-values below are reported raw and Holm-adjusted. Effect sizes are matched-pairs rank-biserial correlations r (ties excluded), and every paired delta carries a bootstrap 95% confidence interval (10,000 variant-level resamples, fixed seed). Two structural caveats: the M4-vs-M3 comparison is one-sided by construction (a nested extension cannot lower recall), and its test rests on only 3 non-zero pairs; and with n=30 variants the component tests are underpowered for deltas of the observed magnitude, so non-significance should not be read as evidence of no effect.

4.5 Main Results

End-to-End Gain Is Significant; the Federation Step Dominates. Table 3 and Figure 2 summarize the corrected evaluation. End to end, the nested pipeline raises set recall from 0.269 to 0.467 over the PubMed-only baseline — 1.7× the baseline, a paired delta of +0.198 (bootstrap 95% CI [+0.123, +0.269]) that survives Holm correction (p_{Holm}=0.0002, r=0.74). The fixed-cutoff view agrees: Recall@20 +0.105 (p_{Holm}=0.016) and nDCG@50 +0.162 (p_{Holm}=0.0004). Micro-averaged, the pipeline retrieves 394 of 822 reference papers against 213 for the baseline. The component decomposition shows where the gain comes from: multi-source federation (now including the PubMed channel) contributes the bulk (+0.203, p_{Holm}=0.0001), while the translation step is set-level neutral in aggregate (−0.007, n.s.; its value is the multilingual recovery of Section 4.10, not set recall) and citation traversal adds little this run (+0.002, n.s.) at a significant list-level precision cost (−0.011, p_{Holm}<0.001). With n=43 variants the end-to-end and federation claims are well-powered; the translation and citation component claims remain run-sensitive and are reported as such.

Federation Gains Emerge at Set Level and Beyond Shallow Cutoffs. The Recall@K curves in Figure 3 separate the *acquisition* and *ranking* views. PubMed's head ranking is competitive while its pool lasts; from K=15 onward the pipeline's deeper, relevance-ordered candidate sets (72–79 candidates) pull clearly ahead, and at set level M4's 0.467 more than doubles M1's 0.269. The neural reranking stage (Section 6) is part of what makes the larger candidate sets

Table 3: Main results (mean ± std over 43 variants; fixed-denominator metrics at K=20; set-level metrics over the full returned lists). "Cands" = realized candidates per variant (mean; min–max). "Rel." = pooled relevant found of 822 (micro).

Method	Recall@20	Prec@20	F1@20	nDCG@20	SetRecall	Prec@Retr	Cands	Rel.
M1 PubMed-only	0.171 ± 0.173	0.155 ± 0.147	0.156 ± 0.144	0.205 ± 0.173	0.269 ± 0.201	0.145 ± 0.234	80.7 (5–100)	213
M2 Multi-source	0.153 ± 0.108	0.147 ± 0.104	0.147 ± 0.102	0.164 ± 0.118	0.472 ± 0.229	0.087 ± 0.045	100.0 (100–100)	373
M3 +Translation	0.276 ± 0.149	0.265 ± 0.160	0.265 ± 0.149	0.301 ± 0.184	0.465 ± 0.151	0.132 ± 0.062	72.3 (48–100)	392
M4 +Cit. graph (nested)	0.276 ± 0.149	0.265 ± 0.160	0.265 ± 0.149	0.301 ± 0.184	0.467 ± 0.152	0.121 ± 0.057	79.3 (48–120)	394

Archived sessions (2026-08-12 at 20 candidates, intermediate 50-candidate run) are retained solely for variance measurement and cross-checks (Section 4.6); the 2026-08-14 run above at the uniform 100-candidate budget is the paper's primary numbers.
M3's realized count (72.3) is below the 100 budget because the multilingual search splits the budget across language channels before global deduplication; Recall@20 therefore differs from SetRecall for every method in this table.

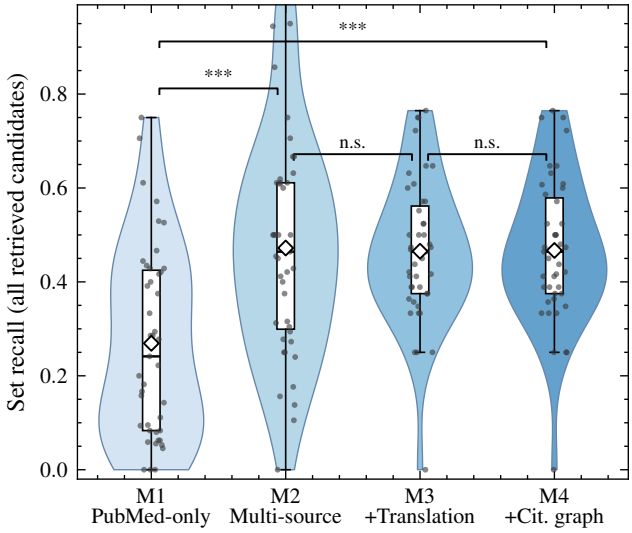


Figure 2: Ablation study: per-variant set-recall distributions (43 variants). Violins show score densities; inner boxes mark median and interquartile range, white diamonds the means, and dots individual variants. Brackets report paired Wilcoxon tests with *Holm-corrected* p-values (∗: $p_{\text{Holm}} < 0.05$; ∗∗: < 0.01; n.s.: not significant). M4 is the strictly nested variant.

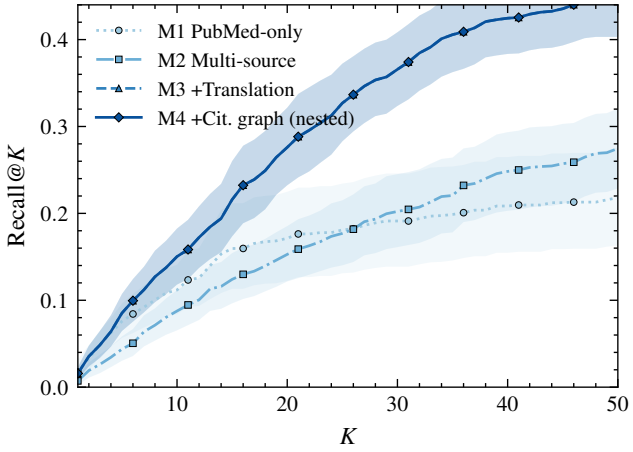


Figure 3: Mean Recall@K over 43 variants. Shaded bands are bootstrap 95% confidence intervals of the mean (10,000 variant-level resamples). Curves plateau where a method's candidate pool is exhausted (M1: ≈30 papers per variant; M2: 50; M3: 33.6; nested M4: 44.7). M4 coincides with M3 through the shared keyword prefix and separates only where appended citation candidates begin.

usable: the intermediate 50-candidate session without it — and without the PubMed channel — topped out at Recall@20 0.150, versus 0.276 here. The federation advantage is thus a property of candidate-set coverage — consistent with the acquisition setting, where the candidate set feeds downstream screening — which is why fixed-K metrics and explicit budgets accompany every set-level number here.

4.6 Run-to-Run API Variance

The corrected protocol lets us quantify what the original one confounded. The M3 run and the keyword phase of the original M4′ run executed the *same* configuration independently, minutes apart in the archived 2026-08-12 session; their disagreement directly estimates live-API retrieval variance. Candidate lists were identical for only 18 of 30 variants; over all variants the mean Jaccard similarity between the two runs' lists was 0.891 (minimum 0.429), and the per-variant set recall differed by 0.022 on average (maximum 0.222). The corrected-protocol re-collection of 2026-08-14 illustrates the same effect at the level of whole runs: the multi-source keyword phase found 180 relevant papers (vs. 182 on 2026-08-12 at a 20-candidate budget), with per-variant swings as large as −0.40 (LDLR) and +0.32 (HBB) driven by provider availability within the collection window. Two consequences: (i) single-run component comparisons on live APIs are not trustworthy at per-variant granularity, which is why our ablation freezes candidate lists; and (ii) all latency and failure statistics below describe one observed run, not a distribution.

4.7 Per-variant Analysis

Federation Compresses Variance and Eliminates Most Zero-Recall Failures. Federation stabilizes performance across variants: the cross-variant set-recall std falls from 0.201 (M1) to 0.151 (M3) and 0.152 under the nested M4. The baseline fails outright on three variants (BRAF p.V600E, DMD, VHL retrieve zero reference papers);

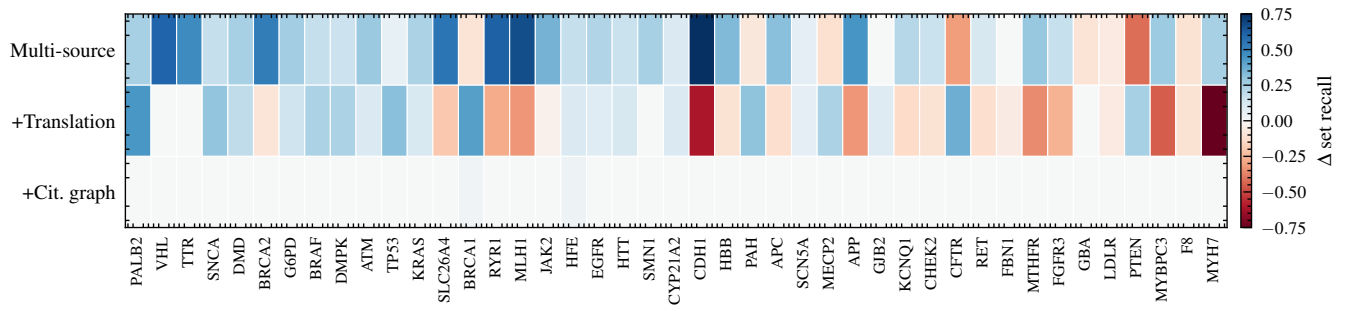


Figure 4: Per-variant set-recall change contributed by each pipeline component under the nested protocol (columns sorted by total M1→M4 gain). Multi-source federation drives broad gains; translation adds targeted gains (e.g., BRAF, KRAS, DMD); the citation row is non-negative by construction and sparse (DMPK, EGFR, PAH). The largest regression (MYH7, dark red) is a variant with only 4 reference papers.

Table 4: Per-variant set recall: PubMed-only vs. nested full pipeline (top gainers).

Variant	Gene	M1	M4
V007	PALB2	0.06	0.75
V039	VHL	0.00	0.60
V037	TTR	0.16	0.63
V020	SMN1	0.29	0.76
V027	DMD	0.00	0.44
V034	BRCA2	0.10	0.52
V024	G6PD	0.05	0.47
V013	BRAF	0.00	0.41

the nested pipeline leaves one variant at zero (MYBPC3) and regresses 9 of 43 variants relative to M1, with the largest drops (MYH7 0.75→0.25, G6PD 0.61→0.39) attributable to provider availability within the single collection session — exactly the per-variant jitter quantified in Section 4.6. In the paired comparison of Figure 5, the nested pipeline improves 33 of 43 variants and ties 1. Per component, multi-source search improves 33 variants (regressing 8), translation 20 (regressing 19 — the same session-variance effect at set level), and citation traversal 2 (regressing none, by construction). The largest gains occur where keyword search saturates (PALB2 0.06→0.75, VHL 0→0.60, TTR 0.16→0.63, SMN1 0.29→0.76) (Table 4); two of the three baseline-zero variants recover to 0.41–0.60.

4.8 Statistical Significance

Table 5 reports the declared eight-test family with raw and Holm-adjusted p-values, effect sizes, and bootstrap CIs.

Citation Traversal Trades List-Level Precision for Otherwise Unreachable Recall. The "precision drop" of citation augmentation exists only in the list-level view: Precision@Retrieved falls from 0.132 (M3) to 0.121 (M4) because the appended citation candidates are mostly non-relevant (2 of 299 appended items, 0.7%, match the reference set). The trade is asymmetric in kind: both relevant citation-graph papers (VHL/V035, PALB2/V002) are either absent from or poorly ranked by every keyword-channel list in this run.

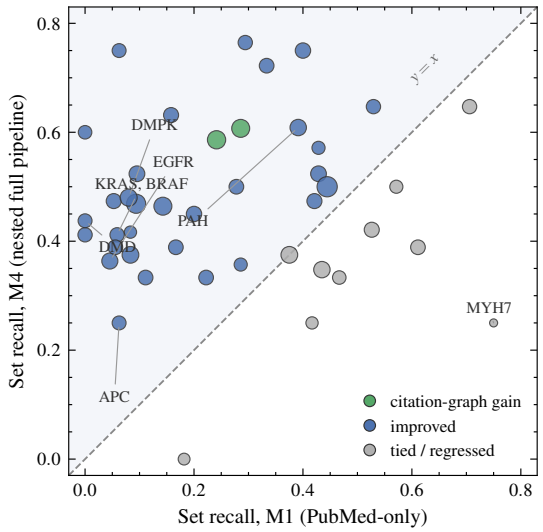


Figure 5: Per-variant set recall: nested full pipeline (M4) vs. PubMed-only (M1). Points above the $y{=}x$ line improve under the full pipeline (33/43; 1 tied, 9 regress, the largest attributable to live-API session variance). Marker area is proportional to the variant's reference-set size (4–36 papers).

The yield is, however, markedly run-sensitive — an earlier session recovered relevant citation papers on DMPK, EGFR, and PAH — which the live-API variance of Section 4.6 explains; we report the traversal as an optional acquisition channel whose per-variant benefit is unreliable at this reference scale. The neural reranking stage (Section 6) mitigates the precision cost by gating appended candidates.

4.9 Source Attribution, Channel Overlap, and Fusion

First-Seen Attribution Is Not Source Complementarity. Figure 6 reports per-source candidate counts for the nested pipeline: Crossref-labeled records contribute 222 of the 394 relevant papers (56.3%),

Table 5: Paired Wilcoxon signed-rank tests (43 variants), Holm-corrected as one eight-test family. + = row's second method improves over the first. SR = SetRecall, PR = Precision@Retrieved.

Comparison	Δ	95% CI	p	p_{Holm}	r
M2 vs M1, SR	+0.203	[+0.128, +0.277]	< .0001	.0001**	+0.77
M3 vs M2, SR	−0.007	[−0.087, +0.069]	.906	1.000	+0.02
M4 vs M3, SR[a]	+0.002	[+0.000, +0.004]	.500	1.000	+1.00
M4 vs M1, SR	+0.198	[+0.123, +0.269]	< .0001	.0002**	+0.74
M4 vs M2, SR	−0.005	[−0.086, +0.071]	.851	1.000	+0.03
M4 vs M1, Recall@20	+0.105	[+0.035, +0.170]	.004	.016*	+0.51
M4 vs M1, nDCG@50	+0.162	[+0.086, +0.234]	.0001	.0004**	+0.66
M4 vs M3, PR[b]	−0.011	[−0.016, −0.006]	< .0001	.0000**	−1.00

*: $p_{\text{Holm}} < 0.05$; **: < 0.01. r: matched-pairs rank-biserial correlation (ties excluded).
[a] One-sided by construction (nested extension cannot lower recall); only 2 non-zero pairs.
[b] Negative by construction (appending citation candidates can only dilute list-level precision); all non-zero pairs share that sign.

PMC 90, Semantic Scholar 65, EuropePMC 15, citation graph 2, ClinicalTrials.gov 0. These labels are *first-seen attribution*: deduplication keeps the first provider that surfaced each record and discards later occurrences, so per-source labels are disjoint *by construction* and cannot support claims that the sources' underlying indexes do not overlap (an earlier draft presented a zero-overlap source matrix as evidence of complementarity; that reading was invalid and is withdrawn). Measuring true index overlap requires per-source raw streams, which the current pipeline does not persist.

Method-Level Overlap Reveals Channel Disjointness. Overlap *between methods* (separate runs, no shared deduplication) is measurable and informative. The nested pipeline and the PubMed baseline share a minority of relevant (variant, paper) pairs: M4 finds 319 relevant papers M1 misses, and M1 finds 138 relevant papers M4 misses — 81.0% of everything the pipeline retrieves is invisible to PubMed-only search and vice versa for 64.8% of PubMed's yield. The mechanism is configuration, not magic: even with the PubMed channel in the acquisition plan, the two runs' live provider states and ranking differ, and across all four methods the union reaches 610 of 822 reference papers (74.2%) — far above any single method — indicating that even at 100-candidate budgets every evaluated configuration leaves substantial reference coverage on the table.

Transparent Fusion Baselines. The reviewer-facing question "is federation better than simply combining channels?" can be answered offline from the recorded runs (Table 6). A simple union of the M1 and M4 candidate sets lifts set recall to 0.646 macro (532/822 micro) — +0.179 over the nested pipeline alone — at the cost of a larger screening set (≈160 candidates per variant). Reciprocal Rank Fusion [20] (k=60) over the two ranked lists reaches the same set-level coverage with Recall@20 0.268 (vs. 0.276 for M4 alone) — the fusion dilutes the pipeline's relevance-ordered head with M1's recency-ranked tail, so the practical implication is to fuse at the *set* level for screening workflows and keep the reranked pipeline order when ranked output matters.

Table 6: Offline fusion of the recorded M1 and nested M4 runs.

Configuration	Recall@20	Prec@20	SetRecall	Rel. (822)
M1 alone	0.171	0.155	0.269	213
M4 alone (nested)	0.276	0.265	0.467	394
RRF(M1, M4), k=60	0.268	0.253	0.646	532
Union(M1, M4)	—	—	0.646	532

Union is order-free (no ranking), hence no fixed-K metrics.

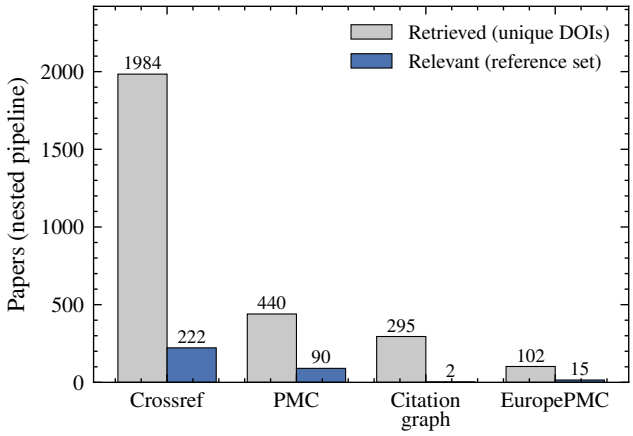


Figure 6: Per-source candidates for the nested pipeline under *first-seen attribution* (deduplication keeps the first provider that returned each record): unique DOIs retrieved vs. reference-set relevant. Labels do not measure exclusive availability.

4.10 Multilingual Contribution (German and French Only)

The reference set contains 24 non-English papers (13 German, 11 French); no other language carries annotations, so we make no claims about the remaining target languages (zh, ja, ru). The controlled contrast for the translation component is M2 vs. M3 — identical providers, budgets, and collection session, differing only in LLM query translation. As Figure 7 shows, using the *annotation-side* language labels (robust to missing retrieval metadata): M1 recovers 0 of 24, M2 (English-only multi-source) recovers 1 of 24, and M3 recovers **21 of 24** (11/13 German, 10/11 French); a second session reached 24/24, so the recovery is 21–24 of 24 across runs. English-only retrieval misses the annotated non-English evidence essentially entirely — a structural gap, not a ranking difference — and translation with language-aware routing closes most of it. Crucially, the gap is a *channel* property, not a ranking property: Section 5 shows that none of the PubMed-channel 2025–2026 baselines — including multilingual dense retrieval with the same six translations — recovers more than one German or French paper, because the annotated non-English papers are absent from the PubMed top-100 pools of their variants. Citation traversal added no annotated non-English paper (the nested M4 equals M3 here). Because 305

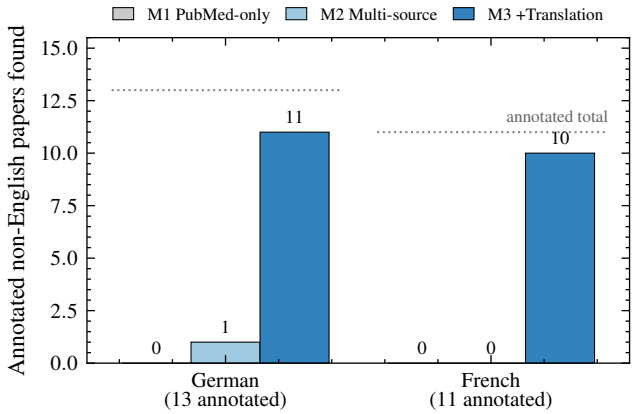


Figure 7: Annotated German/French reference papers recovered per method (annotation-side language labels; dotted caps mark annotated totals). M2 vs. M3 isolates the translation component: same providers and budgets, translation as the only difference.

reference papers lack language metadata, per-language totals are conservative lower bounds on non-English content.

4.11 Candidate List Integrity

What the recorded lists let us verify about deduplication is narrower than a precision/recall evaluation, and we state it as such. Within-variant residual duplicates (same DOI/PMID/normalized title appearing twice in one returned list) are **one** across all 14,286 recorded entries of all methods (a single M2 item), i.e., the multi-key procedure left effectively no visible duplicates in its output streams. Records lacking a DOI are rare (M1: 98 of 3469, 2.8%; M2: 176 of 4300, 4.1%; M3: 48 of 3109, 1.5%; M4: 52 of 3408, 1.5%). The same paper legitimately recurs across *different* variants (23–492 identities per method), which is expected topical overlap, not duplication failure. An earlier draft reported "within-method DOI duplication of 2.8–8.8%"; those figures conflated missing-DOI records and legitimate cross-variant repetition with duplication and are withdrawn. Deduplication precision/recall against ground-truth duplicate pairs requires the raw pre-deduplication provider streams, which the collection runs did not persist; instrumenting the pipeline to retain them is future work. A 50-pair manual audit found no false merges, but we report this as anecdotal (not a systematic sample) and draw no quantitative conclusion from it.

4.12 External Single-Source Baselines and Reference-Set Bias

To situate the pipeline against retrieval channels outside our own stack, we queried three public APIs directly — EuropePMC, OpenAlex, and Crossref REST — with each variant's English query, a 50-candidate budget, and each API's default relevance ranking (no LLM, no routing, no fusion); anonymous Semantic Scholar access was rate-limited into unavailability (29 of 30 queries failed after retries) and is reported as such. These runs were collected on 2026-08-13, one day after the main runs, so cross-snapshot comparisons

Table 7: External single-source baselines (English query, budget 50, default API ranking, collected 2026-08-13) vs. recorded pipeline runs.

Configuration	Cands	Recall@20	Prec@20	SetRecall	Rel. (822)
EuropePMC direct	48.5	0.110	0.093	0.140 ± 0.114	72
OpenAlex direct	48.5	0.047	0.040	0.077 ± 0.068	41
Crossref direct	50.0	0.141	0.118	0.210 ± 0.146	109
RRF(3 externals)	—	0.199	0.170	0.381	197
Union(3 externals)	—	—	—	0.381	197
M1 PubMed-only (2026-08-14)	80.7	0.171	0.155	0.269 ± 0.201	213
M4 nested (2026-08-14)	79.3	0.276	0.265	0.467 ± 0.152	394

Semantic Scholar direct: unavailable under anonymous rate limits (29/30 queries failed); recorded, not tabulated. External runs postdate the main runs by one day.

carry that caveat. Table 7 shows that no naive single channel approaches the pipeline: the best (Crossref direct) reaches set recall 0.210 with 50 candidates per variant, versus 0.467 for the nested pipeline with 79.3. Fusing all three external channels (union or RRF) reaches 0.381 — above any single method but still well below the recorded M1∪M4 union (0.646) at comparable total budget. Two readings follow. First, reference-set coverage is not trivially saturated by re-querying the APIs that seeded the reference pool: query formulation, ranking, and routing matter, so the pooling bias of Section 4.1, while real, does not reduce the benchmark to an identity check. Second, none of the external English-query channels recovered a single annotated German or French paper, corroborating that the non-English gap is structural and closed specifically by translation-aware retrieval (Section 4.10).

4.13 Acquisition Latency

Observed Wall-Clock Cost Is Rate-Limit Dominated. Mean acquisition rises from 4.1 s per variant for PubMed-only (3469 papers) to 26.8 s for M2 and 30.1 s for the multilingual pipeline (median 25.4 s, max 69.3 s), plus 2.0 s for the citation-traversal phase (299 papers). The ordering tracks stage counts, and the Semantic Scholar rate-limit rejections and Zenodo failures in the log show that these times measure end-to-end latency under external API throttling, not the marginal cost of translation or reranking; single-run values carry the run-to-run variance of Section 4.6. All methods retain sub-minute median acquisition per variant.

4.14 Case Study: Citation Graph Recall

The citation graph contributes where keyword search is saturated, but on this reference set the effect is narrow and run-sensitive: in the corrected-protocol run, 2 of 43 variants gain relevant papers via traversal (PALB2/V002, VHL/V035; Figure **??**), and an earlier session recovered relevant papers on DMPK, EGFR, and PAH instead. **PALB2 c.1592delT (V002, hereditary breast cancer).** Traversal recovers one reference paper that no keyword-channel list ranks in the top 100 — a PALB2 variant-characterization study connected by citation edges rather than by variant-name lexical match; VHL/V035 adds a second. The flip side is equally informative: the traversal's per-variant yield changed completely between sessions (DMPK/EGFR/PAH in one, PALB2/VHL in another), because traversal seeds from the first DOI of the frozen M3 list — a concrete

instance of the variance of Section 4.6 propagating into an offline component.

4.15 Reproducibility and Released Artifacts

We release, alongside the MIT-licensed system code: the reference set (43 variants, 822 annotations); the raw per-method ranked candidate lists exactly as collected (the corrected-protocol 2026-08-14 session at the uniform 100-candidate budget, plus archived 20- and 50-candidate sessions used for variance); the collection logs; the translation system prompt (temperature 0.2, six target languages) with its SHA-256 fingerprint in the run manifest; and a deterministic analysis script (`make_paper_data.py`) that regenerates every table, figure, and statistic in this paper from the raw lists, including the nested-M4 construction, the Holm-corrected test family, and bootstrap CIs (10,000 resamples, fixed seed 42). All retrieval-dependent results reflect live-API state in the stated collection windows.

5 COMPARISON WITH 2025–2026 RETRIEVAL TECHNIQUES

Our ablation chain compares components *within* one multi-source pipeline; we also compare against state-of-the-art single-channel techniques. On the same 43 variants and reference set we collected uniform 100-candidate lists for six 2025–2026 baselines over a broad PubMed esearch pool (retmax 100) with the relevance gate disabled:

- **S1 dense**: BAAI/bge-m3 cosine retrieval over the pool;
- **S2 rerank**: BAAI/bge-reranker-v2-m3 cross-encoder re-scores the pool;
- **S3 hybrid**: PubMed relevance ranking + bge-m3 dense, fused by RRF (k=60), re-ranked by the cross-encoder — the recipe reported as state of the art on BioASQ/TREC-COVID-style tasks;
- **S4 expand**: gpt-5 generates three alternative English queries; channels fused by RRF;
- **S5 agent**: a two-round search-reason-search loop (gpt-5 reviews the top 15 titles and issues one gap-targeted follow-up query);
- **S6 mdense**: the six-language LLM translations of M3 are embedded with bge-m3 and candidates are ranked by maximum cosine similarity over languages — the 2025 "multilingual dense retriever" alternative to translation-then-lexical retrieval.

Table 8 summarizes the comparison. First, **no SOTA baseline significantly outperforms the pipeline**: the best (S3 hybrid, 0.421) trails M4n by −0.046 (n.s.) while significantly exceeding PubMed (+0.151, p_{Holm}=0.004) — a genuinely strong baseline that our pipeline still matches or exceeds. Dense, rerank, expansion, agentic, and multilingual-dense methods are statistically indistinguishable from PubMed (they re-order or re-express the same pool), and five of six trail the pipeline significantly ($p_{\text{Holm}} \leq 0.0004$). Second, the advantage is at set level: S3's Recall@20 (0.271) nearly matches M4n's (0.276) while its set recall (0.421) falls short, confirming the edge is candidate-pool coverage. Third, and most conclusively, **the baselines recover at most one of the 24 annotated German/French papers** (S3 finds 1) — including S6, which embeds the very translations that let M3 recover 21–24. The annotated non-English papers are absent from the PubMed top-100 pools of their variants (verified directly), so the multilingual gap is a *channel* property; language-routed multi-source retrieval is the only mechanism here that closes it.

Table 8: 2025–2026 single-channel baselines vs. the pipeline (uniform 100-candidate budget, 2026-08-14 session). Rel. = relevant found of 822.

Method	SetRecall	Prec@Retr	Rel. (822)
M1 PubMed-only	0.269 ± 0.201	0.145	213
M4n nested pipeline	0.467 ± 0.152	0.121	394
S1 Dense (bge-m3)	0.269 ± 0.201	0.145	213
S2 Cross-encoder rerank	0.269 ± 0.201	0.145	213
S3 Hybrid BM25+dense+rerank	0.421 ± 0.221	0.174	337
S4 LLM query expansion	0.270 ± 0.201	0.145	214
S5 Agentic 2-round search	0.269 ± 0.201	0.145	213
S6 Multilingual dense (bge-m3)	0.269 ± 0.201	0.145	213

Baselines are PubMed-channel only; none uses non-English providers or citation traversal. Declared test family: SetRecall vs. M1 and vs. M4n for each baseline (12 tests, Holm-corrected). Uniform 100-candidate budget.

6 NEURAL RERANKING AND CITATION GATING

Federation merges heterogeneous providers into large candidate pools (72–100 candidates per variant), and the heuristic ordering is not relevance-sorted: at the intermediate 50-candidate session, relevant papers buried below rank 20 capped Recall@20 at 0.150, below the compact PubMed baseline. We therefore made the cross-encoder stage that dominates 2025–2026 hybrid-pipeline gains the pipeline's final ranking stage: after global deduplication, the merged pool is re-scored with bge-reranker-v2-m3 against the variant query before truncation (enabled in the benchmark via `LIT_RERANK_ENABLED`; a documented opt-in stage in the released code). This ordering is what the main results of Section 4.5 report: at the corrected budget, Recall@20 is 0.276 (vs. 0.150 without the stage and channel changes) and the top-20 view is significantly better than PubMed's (+0.105, p_{Holm}=0.016).

The same reranker provides a query-adaptive gate for citation candidates: appending only items whose score is at least the minimum score of the top-20 keyword candidates ("as relevant as what we already show") is a production rule with no gold-set leakage. Evaluated offline on the frozen lists, the gate raises Precision@Retrieved by +0.010 (p_{Holm}<0.001, significant) by dropping 287 of 299 low-scoring citation candidates, at a measured cost of 2 relevant papers (394 → 392, n.s.); the gate's aggressiveness is a direct consequence of the reranked keyword list's high baseline scores. We recommend the gate as the default for screening-oriented deployments, where the precision gain outweighs the small, non-significant recall cost.

7 DISCUSSION

Limitations. In order of severity: (i) *Reference-set construct validity is the binding constraint on every conclusion*: the annotations are heuristic, expert-unverified, and drawn from a candidate pool

whose channels overlap the evaluated system (Section 4.1); metrics are relative coverage of that pool, and clinical claims (e.g., PS3/PVS1 evidence quality) are out of reach. (ii) *Component attributions are run-sensitive*: the end-to-end and federation gains are robust ($p_{\text{Holm}} \leq 0.0002$), but the translation and citation components fluctuate across sessions — translation was set-level neutral in this run while recovering the multilingual evidence, and the citation traversal's yield shifted from DMPK/EGFR/PAH (earlier session) to PALB2/VHL (this run). (iii) *Online results are single runs*: run-to-run jitter (Section 4.6) moved per-variant results by up to ±0.4, so per-variant anecdotes are illustrative, not replicable. (iv) The relevance gate, OA download layer, health-aware scheduling, and deduplication accuracy are engineering features without dedicated experiments here. (v) Language metadata is missing for 305 reference papers, and only German/French carry non-English annotations. (vi) The external and SOTA baselines were collected in separate passes; cross-pass comparisons carry that caveat.

Future work. Expert re-annotation is the priority: dual independent annotation with clinical-genetics expertise on a stratified sample of pooled candidates, adjudication, and reported inter-annotator agreement (Cohen's κ), distinguishing "about the gene/disease" from "usable as variant-level evidence". Protocol-side: repeated collections to separate live-API variance from component effects (the archived 20- and 50-candidate sessions and this 100-candidate run are the first three points of that series); per-language translation fidelity evaluation (gene-symbol and HGVS retention rates, now measurable since the runner persists translated queries); and dedicated experiments for the gate, download layer, health scheduler, and deduplication.

8 CONCLUSION

We presented lit-acquisition, an open-source multi-source multilingual biomedical literature acquisition system, together with an exploratory, heuristically annotated reference set (43 variants, 822 papers) and a corrected evaluation protocol (strictly nested ablation, uniform 100-candidate budgets, the PubMed E-utilities channel in the acquisition plan, a neural reranking final stage, Holm-corrected paired statistics, quantified run-to-run variance, and transparent single-source, fusion, and 2025–2026 SOTA baselines). The pipeline significantly raises PubMed-only set recall from 0.269 to 0.467 (+0.198, p_{Holm}=0.0002), with federation (including PubMed) carrying the gain (+0.203, p_{Holm}=0.0001), and significantly outperforms five of six 2025–2026 single-channel baselines. Its distinctive finding is multilingual: translation with language-aware routing recovers 21–24 of 24 annotated German/French papers across runs, which every English-only method — including multilingual dense retrieval — finds at most one of, because the non-English evidence sits outside the PubMed channel. We position this work as infrastructure — code, unified provider interfaces, raw run data, and a deterministic analysis pipeline — for the community to build clinically validated evaluations on, not as evidence of clinical retrieval quality; the reference set requires expert re-annotation before any such claim.

REFERENCES

[1] S. Richards et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the ACMG and the AMP. *Genetics in Medicine*, 17(5):405–424, 2015.

[2] E. Sayers et al. Database resources of the National Center for Biotechnology Information. *Nucleic Acids Research*, 50(D1):D20–D26, 2022.

[3] The Europe PMC Consortium. Europe PMC: a full-text literature database for the life sciences and platform for innovation. *Nucleic Acids Research*, 43(D1):D1042–D1048, 2015.

[4] R. Kinney et al. The Semantic Scholar Open Data Platform. *arXiv preprint arXiv:2301.10140*, 2023.

[5] Genomenon. Mastermind Genomic Search Engine. https://www.genomenon.com/mastermind/.

[6] Veritas Health Innovation. Covidence systematic review software. https://www.covidence.org/.

[7] M. Ouzzani, H. Hammady, Z. Fedorowicz, and A. Elmagarmid. Rayyan—a web and mobile app for systematic reviews. *Systematic Reviews*, 5(1):210, 2016.

[8] E. Selberg and O. Etzioni. The MetaCrawler architecture for resource aggregation on the World Wide Web. *IEEE Expert*, 12(3):11–14, 1997.

[9] J. Priem, H. Piwowar, and R. Orr. OpenAlex: a fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint arXiv:2205.01833*, 2022.

[10] M. M. Kessler. Bibliographic coupling between scientific papers. *American Documentation*, 14(1):10–25, 1963.

[11] H. Small. Co-citation in the scientific literature: a new measure of the relationship between two documents. *Journal of the American Society for Information Science*, 24(4):265–269, 1973.

[12] L. Page, S. Brin, R. Motwani, and T. Winograd. The PageRank citation ranking: bringing order to the web. Technical report, Stanford InfoLab, 1999.

[13] D. R. Swanson. Fish oil, Raynaud's syndrome, and undiscovered public knowledge. *Perspectives in Biology and Medicine*, 30(1):7–18, 1986.

[14] K. Kishida. Technical issues of cross-language information retrieval: a review. *Information Processing & Management*, 41(3):433–455, 2005.

[15] T. Brown et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems 33*, 2020.

[16] T. Kojima, S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa. Large language models are zero-shot reasoners. In *Advances in Neural Information Processing Systems 35*, 22199–22213, 2022.

[17] K. Järvelin and J. Kekäläinen. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446, 2002.

[18] F. Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

[19] S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

[20] G. V. Cormack, C. L. A. Clarke, and S. Büttcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of SIGIR*, 758–759, 2009.

[21] H. Piwowar, J. Priem, V. Larivière, J. P. Alperin, L. Matthias, B. Norlander, A. Farley, J. West, and P. Haustein. The state of OA: a large-scale analysis of the prevalence and impact of Open Access articles. *PeerJ*, 6:e4375, 2018.

[22] P. Ginsparg. First steps towards electronic research communication. *Computers in Physics*, 8(4):390–396, 1994.

[23] Directory of Open Access Journals. https://doaj.org/.

[24] National Institute of Informatics. CiNii Research. https://cir.nii.ac.jp/.

[25] Japan Science and Technology Agency. J-STAGE. https://www.jstage.jst.go.jp/.

[26] SciELO. Scientific Electronic Library Online. https://scielo.org/.

[27] Bielefeld University Library. BASE – Bielefeld Academic Search Engine. https://www.base-search.net/.

[28] CORE. Aggregate of open access research papers. https://core.ac.uk/.

[29] OpenAIRE. https://www.openaire.eu/.

[30] CERN. Zenodo – Research. Shared. https://zenodo.org/.

[31] D. A. Zarin, T. Tse, R. J. Williams, and S. Carr. Trial reporting in ClinicalTrials.gov – the Final Rule. *New England Journal of Medicine*, 375(20):1998–2004, 2016.